

Q.1 Consider the operator precedence and associativity rules for the *integer* arithmetic operators given in the table below.

Operator	Precedence	Associativity
+	Highest	Left
–	High	Right
*	Medium	Right
/	Low	Right

The value of the expression as per the above rules is _____

GATE CSE 2024 Set 1

Answer: 6

Q.2 Consider the following sequence of operations on an empty stack.

push(54); push(52); pop(); push(55); push(62); s = pop();

Consider the following sequence of operations on an empty queue.

enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();

The value of s + q is _____

GATE CSE 2021 Set 1

Answer: 86

Q.3 A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is correct (n refers to the number of items in the queue)?

GATE CSE 2016 Set 1

- ☒ A Both operations can be performed in $O(1)$ time
- ☐ B At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$
- ☐ C The worst case time complexity for both operations will be $\Omega(n)$
- ☐ D Worst case time complexity for both operations will be $\Omega(\log n)$

Answer: (A) Both operations can be performed in $O(1)$ time

Q.4 The result evaluating the postfix expression 10 5 + 60 6 / * 8 – is

GATE CSE 2015 Set 3

☐ A 284

☐ B 213

☐ C 142

☐ D 71

Answer: (C) 142

Q.5 A function f defined on stacks of integers satisfies the following properties. $f(\emptyset) = 0$ and $f(\text{push}(S, i)) = \max(f(S), 0) + i$ for all stacks S and integers i . If a stack S contains the integers 2, -3, 2, -1, 2 in order from bottom to top, what is $f(S)$? **GATE CSE 2005**

☐ A 6

☐ B 4

☐ C 3

☐ D 2

Answer: (C) 3

Q.6 The best data structure to check whether an arithmetic expression has balanced parentheses is a **GATE CSE 2004**

☐ A queue

☐ B stack

☐ C tree

☐ D list

Answer: (B) stack

Q.7 Which of the following is essential for converting an infix expression to the postfix form efficiently? **GATE CSE 1997**

- ☐ A An operator stack
- ☐ B An operand stack
- ☐ C An operand stack and an operator stack
- ☐ D A parse tree

Answer: (A) An operator stack

Q.8 Consider the following statements:

- (i) First-in-first out types of computations are efficiently supported by STACKS.
 - (ii) Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.
 - (iii) Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.
 - (iv) Last-in-first-out type of computations are efficiently supported by QUEUES.
- GATE CSE 1996**

- ☐ A (ii) and (iii) are true
- ☐ B (i) and (ii) are true
- ☐ C (iii) and (iv) are true
- ☐ D (ii) and (iv) are true

Answer: (A) (ii) and (iii) are true

Q.9 Consider a stack S and a queue Q. Both of them are initially empty and have the capacity to store ten elements each. The elements 1, 2, 3, 4, and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeuing all elements. The output obtained by following this process is 43125.

Given the output, the objective is to predict whether an element was assigned to S or Q. Which of the following options is/are possible valid assignment(s) of the elements?

Note: In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q.

GATE CSE 2026 Set 2

A 1S, 2Q, 3S, 4S, 5Q

B 1Q, 2Q, 3S, 4S, 5Q

C 1Q, 2Q, 3Q, 4S, 5S

D 1S, 2S, 3S, 4Q, 5Q

Answer: (A) 1S, 2Q, 3S, 4S, 5Q

(B) 1Q, 2Q, 3S, 4S, 5Q

Q.10 Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack

1. without deleting the corresponding record, and
2. without increasing the complexities of the standard stack operations.

Which one or more of the following approach(es) can achieve it?

GATE CSE 2025 Set 2

A Keep with every record in the stack, a pointer to the record with the smallest key below it.

B Keep a pointer to the record with the smallest key in the stack.

C Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.

D Keep a Min-Heap in which the key values of the records in the stack are maintained.

Answer: (A) Keep with every record in the stack, a pointer to the record with the smallest key below it.

Q.11 Let **S1** and **S2** be two stacks. **S1** has capacity of 4 elements. **S2** has capacity of 2 elements. **S1** already has 4 elements: 100, 200, 300, and 400, whereas **S2** is empty, as shown below.

Stack S1

400 (Top)

Stack S1
300
200
100
Stack S2

Only the following three operations are available:

PushToS2: Pop the top element from **S1** and push it on **S2**.

PushToS1: Pop the top element from **S2** and push it on **S1**.

GenerateOutput: Pop the top element from **S1** and output it to the user.

Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.

GATE CSE 2024 Set 2

Which of the following output sequences can be generated by using the above operations?

A 100, 200, 400, 300

B 200, 300, 400, 100

C 400, 200, 100, 300

D 300, 200, 400, 100

Answer: (B) 200, 300, 400, 100

(C) 400, 200, 100, 300

(D) 300, 200, 400, 100

Q.12 Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q , both of which are initially empty.

GATE CSE 2023

I: push the elements of a from a_0 to a_5 in that order into S .

II: enqueue the elements of a from a_0 to a_5 in that order into Q .

III: pop an element from S.

IV: dequeue an element from Q.

V: pop an element from S.

VI: dequeue an element from Q.

VII: dequeue an element from Q and push the same

VIII: Repeat operation VII three times.

IX: pop an element from S.

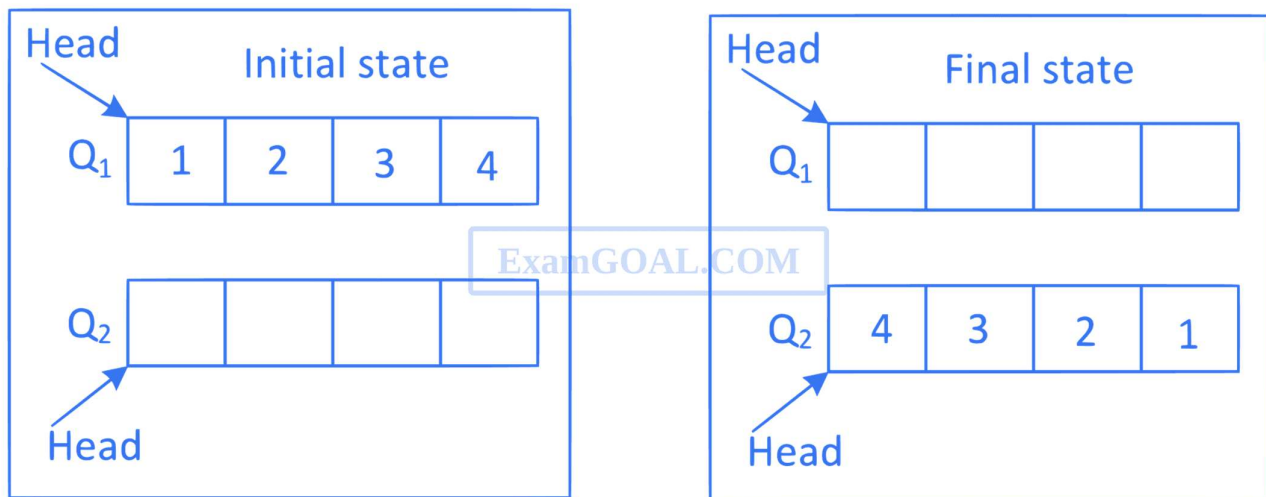
X: pop an element from S.

The top element of S after executing the above operations is _____.

Answer: 8

Q.13 Consider the queues Q_1 containing four elements and Q_2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are Enqueue (Q, element) and Dequeue (Q). The minimum number of Enqueue operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order (shown as the Final State in the figure) without using any additional storage is _____.

GATE CSE 2022



Your input _____

Answer: 0

Q.14 Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

GATE CSE 2014 Set 2

- A** A queue cannot be implemented using this stack.
- B** A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.
- C** A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.
- D** A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each

Answer: (C) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.

Q.15 Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, $\text{REAR} = \text{FRONT} = 0$. The conditions to detect queue full and queue empty are

GATE CSE 2012

- full: $(\text{REAR} + 1)$
A $\text{mod } n == \text{FRONT}$
empty: $\text{REAR} == \text{FRONT}$

- full: $(\text{REAR} + 1)$
B $\text{mod } n == \text{FRONT}$
empty: $(\text{FRONT} + 1)$

- full: $\text{REAR} == \text{FRONT}$
C empty: $(\text{REAR} + 1)$
 $\text{mod } n == \text{FRONT}$

- full: $(\text{FRONT} + 1)$
D $\text{mod } n == \text{REAR}$
empty: $\text{REAR} == \text{FRONT}$

Answer: (A) full: $(\text{REAR} + 1)$

$\text{mod } n == \text{FRONT}$

empty: $\text{REAR} == \text{FRONT}$

Q.16 Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:

- i. isEmpty (Q) — returns true if the queue is empty, false otherwise.
- ii. delete (Q) — deletes the element at the front of the queue and returns its value.
- iii. insert (Q, i) — inserts the integer i at the rear of the queue.

Consider the following function:

```
void f (queue Q) {  
    int i;  
    if (!isEmpty(Q)) {  
        i = delete(Q);  
        f(Q);  
        insert(Q, i);  
    }  
}
```

What operation is performed by the above function f ?

GATE CSE 2007

- ☐ A Leaves the queue Q unchanged
- ☐ B Reverses the order of the elements in the queue Q
- ☐ C Deletes the element at the front of the queue Q and inserts it at the rear keeping the other elements in the same order
- ☐ D Empties the queue Q

Answer: (B) Reverses the order of the elements in the queue Q

Q.17 The following postfix expression with single digit operands is evaluated using a stack: $8\ 2\ 3\ ^\wedge\ /\ 2\ 3\ *\ +\ 5\ 1\ *\ -$ Note that $^\wedge$ is the exponentiation operator. The top two elements of the stack after the first $*$ is evaluated are:

GATE CSE 2007

A 6, 1

B 5, 7

C 3, 2

D 1, 5

Answer: (A) 6, 1

Q.18 Consider the following C program:

GATE CSE 2007

```
#include
#define EOF -1

void push (int); /* push the argument on the stack */
int pop (void); /* pop the top of the stack */
void flagError ();

int main () {
    int c, m, n, r;
    while ((c = getchar ()) != EOF) {
        if (isdigit (c) )
            push (c);
        else if ((c == '+') || (c == '*')) {
            m = pop ();
            n = pop ();
            r = (c == '+') ? n + m : n*m;
            push (r);
        } else if (c != ' ')
            flagError ();
    }
    printf("%c", pop ());
}
```

What is the output of the program for the following input ? 5 2 * 3 3 2 + * +

A 15

B 25

C 30

D 150

Answer: (B) 25

Q.19 The following function computes the value of mC_n correctly for all legal values m and n ($m \geq 1, n \geq 0$ and $m > n$)

GATE CSE 2006

```
int func(int m, int n)
{
    if (E) return 1;
    else return(func(m - 1, n) + func(m - 1, n - 1));
}
```

In the above function, which of the following is the correct expression for E?

A $(n == 0) \parallel (m == 1)$

B $(n == 0) \&\& (m == 1)$

C $(n == 0) \parallel (m == n)$

D $(n == 0) \&\& (m == n)$

Answer: (C) $(n == 0) \parallel (m == n)$

Q.20 An implementation of a queue Q, using two stacks S1 and S2, is given below :

```
void insert(Q, X){
    push(S1, X);
}
```

```

void delete(Q){
if(stack - empty(S2)) then {
    if(stack - empty(S1)) then {
        print("Q is empty");
        return;
    }else while (!(stack - empty(S1))){
        X = pop(S1);
        push(S2, X);
    }
    X = pop(S2);
}
}

```

Let n insert and $m(\leq n)$ delete operations be performed in an arbitrary order on an empty queue Q . Let x and y be the number of push and pop operations performed respectively in the process. Which one of the following is true for all m and n ? **GATE CSE 2006**

A $n + m \leq x \leq 2n$ and $2m \leq y \leq n + m$

B $n + m \leq x \leq 2n$ and $2m \leq y \leq 2n$

C $2m \leq x \leq 2n$ and $2m \leq y \leq n + m$

D $2m \leq x \leq 2n$ and $2m \leq y \leq 2n$

Answer: (A) $n + m \leq x \leq 2n$ and $2m \leq y \leq n + m$

Q.21 A program attempts to generate as many permutation as possible of the string “abcd” by pushing the character a,b,c,d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following a strings CANNOT be generated using this program? **GATE CSE 2004**

A abcd

B dcba

C cbad

D cabd

Answer: (D) cabd

Q.22 Assume that the operators $+$, $-$, $\times$, are left associative and $^$ is right associative. The order of precedence (from highest to lowest) is $^$, $\times$, $+$, $-$. The postfix expression corresponding to the infix expression $a + b \times c - d^e^f$ is

GATE CSE 2004

A $abc \times + def^{^^}-$

B $abc \times + de^f^--$

C $ab + c \times d - e^f^--$

D $- + a \times bc^{^^}def$

Answer: (A) $abc^{\times} + def^{^^} -$

Q.23 Let S be a stack of size $n \geq 1$. Starting with the empty stack, suppose we push the first n natural numbers in sequence, and then perform n pop operations. Assume that Push and Pop operations take X seconds each, and Y seconds elapse between the end of one such stack operation and the start of the next operation. For $m \geq 1$, define the stack-life of m as the time elapsed from the end of Push(m) to the start of the pop operation that removes m from S . The average stack-life of an element of this stack is

GATE CSE 2003

A $n(X + Y)$

B $3Y + 2X$

C $n(X + Y) - X$

D $Y + 2X$

Answer: (C) $n(X + Y) - X$

Q.24 Compute the post fix equivalent of the following expression. $3 * \log(x+1) - a/2$ **GATE CSE 1998**

Answer: $3x1 + \log*a2/-$

Q.25 What value would the following function return for the input $x = 95$?

GATE CSE 1998

function fun (x:integer):integer;

Begin

If $x > 100$ then fun : $x - 10$

Else fun : fun(fun (x + 11))

End;

A 89

B 90

C 91

D 92

Answer: (C) 91

Q.26 A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented INSERT (Q, C, K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETETMIN(Q). For a sequence of operations, the keys chosen are in **GATE CSE 1997**

☐ A non-increasing order

☐ B non-decreasing order

☐ C strictly increasing order

☐ D strictly decreasing order

Answer: (A) non-increasing order

Q.27 The postfix expression for the infix expression $A + B * (C + D) / F + D * E$ is: **GATE CSE 1995**

☐ A $AB + CD + *F/D + E*$

☐ B $ABCD + *F/+DE*+$

☐ C $A * B + CD / F * DE ++$

☐ D $A + *BCD / F * DE ++$

Answer: (B) $ABCD + *F/+DE*+$

Q.28 Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order? **GATE CSE 1994**

☐ A 3, 4, 5, 1, 2

☐ B 3, 4, 5, 2, 1

☐ C 1, 5, 2, 3, 4

☐ D 5, 4, 3, 1, 2

Answer: 3, 4, 5, 2, 1

Q.29 The following sequence of operations is performed on stack: PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP. The sequence of the value popped out is:

GATE CSE 1991

A 20,10,20,10,20

B 20,20,10,10,20

C 10,20,20,10,20

D 20,20,10,20,10

Answer: (B) 20, 20, 10, 10, 20

Q.30 Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list.

Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?

GATE CSE 2023

A SLLdel is $O(1)$ and DLLdel is $O(n)$

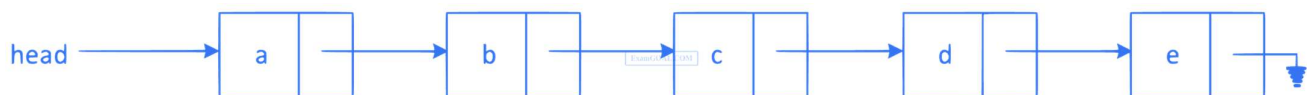
B Both SLLdel and DLLdel are $O(\log(n))$

C Both SLLdel and DLLdel are $O(1)$

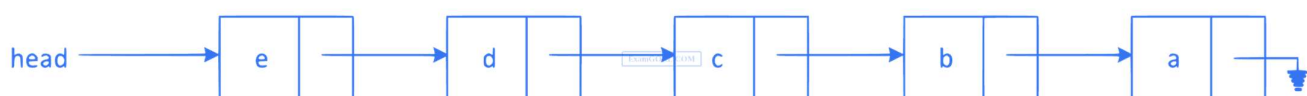
D SLLdel is $O(n)$ and DLLdel is $O(1)$

Answer: (D) SLLdel is $O(n)$ and DLLdel is $O(1)$

Q.31 Consider the problem of reversing a singly linked list. To take an example, given the linked list below:



the reversed linked list should look like



Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?

GATE CSE 2022

- A** The best algorithm for the problem takes $\theta(n)$ time in the worst case.
- B** The best algorithm for the problem takes $\theta(n \log n)$ time in the worst case.
- C** The best algorithm for the problem takes $\theta(n^2)$ time in the worst case.
- D** It is not possible to reverse a singly linked list in $O(1)$ space.

Answer: (A) The best algorithm for the problem takes $\theta(n)$ time in the worst case.

Q.32 What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

GATE CSE 2020

- A** $\Theta(n^2)$
- B** $\Theta(n)$
- C** $\Theta(n \log n)$
- D** $\Theta(1)$

Answer: (A) $\theta(n^2)$

Q.33 N items are stored in a sorted doubly linked list. For a delete operation, a pointer is provided to the record to be deleted. For a decrease-key operation, a pointer is provided to the record on which the operation is to be performed.

An algorithm performs the following operations on the list in this order:

$\theta(N)$, delete, $O(\log N)$ insert, $O(\log N)$ find, and $\theta(N)$ decrease-key. What is the time complexity of all these operations put together?

GATE CSE 2016 Set 2

A $O(\log^2 N)$

B $O(N)$

C $O(N^2)$

D $\Theta(N^2 \log N)$

Answer: (C) $O(N^2)$

Q.34 Let P be a singly linked list, Let Q be the pointer to an intermediate node x in the list. What is the worst-case time complexity of the best known algorithm to delete the node x from the list?

GATE CSE 2004

A $O(n)$

B $O(\log^2 n)$

C $O(\log n)$

D $O(1)$

Answer: (D) $O(1)$

Q.35 In the worst case, the number of comparisons needed to search a singly linked list of length n for a given element is

GATE CSE 2002

A $\log_2 n$

B $n/2$

C $\log_2 n - 1$

D n

Answer: (D) n

Q.36 Consider the following code snippet in C language that computes the number of nodes in a non-empty singly linked list pointed to by the pointer variable head.

GATE CSE 2026 Set 1

```

struct node{

    int elt;

    struct node *next;

};

int getListSize (struct node *head)

{

    if( E1 ) return 1;

    return E2;

}

```

Which one of the following options gives the correct replacements for the expressions E 1 and E 2 ?

- ☒ A E1: head == NULL
E2: 1 + getListSize(head)

- ☒ B E1: head → next == NULL
E2: 1 + getListSize(head → next)

- ☒ C E1: head == NULL
E2: 1 + getListSize(head → next)

- ☒ D E1: head → next == NULL
E2: 1 + getListSize(head)

Answer: (B) E1: head-> next == NULL

E2: 1+ getListSize(head->size)

Q.37 Let LIST be a datatype for an implementation of linked list defined as follows:

GATE CSE 2025 Set 1

```

typedef struct list {

    int data;

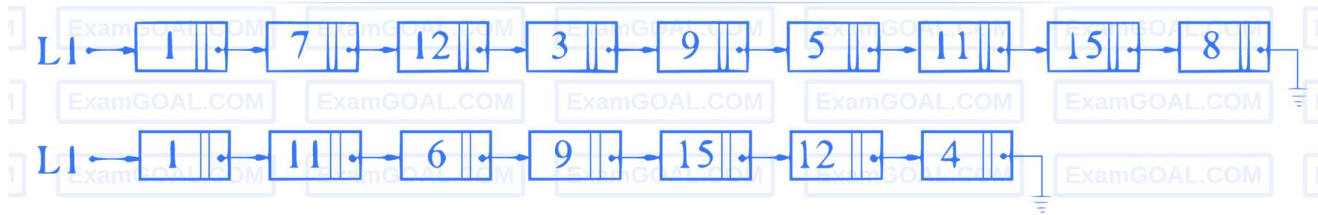
    struct list *next;

}

```

} LIST;

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes. Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____. (Answer in integer)



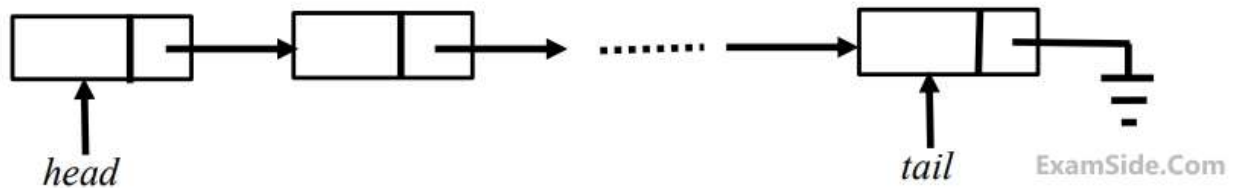
```
int find (int query, LIST *list) {
    while (list != NULL) {
        if(list -> data == query) return 1;
        list = list -> next;
    }
    return 0;
```

```
int main() {
    ... ..
    ptr1=L1; ptr2=L2;
    while (ptr1 -> next != NULL) {
        query = ptr1 -> next -> data;
        if (find (query, L2))
            ptr1 -> next = ptr1 -> next -> next;
        else ptr1 = ptr1 -> next;
    }
    ... ..
    return 0;
```

Your input ____

Answer: 5

Q.38 A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure? **GATE CSE 2018**

A $\theta(1), \theta(1)$

B $\theta(1), \theta(n)$

C $\theta(n), \theta(1)$

D $\theta(n), \theta(n)$

Answer: (B) $\theta(1), \theta(n)$

Q.39 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42.

Which one of the following is the postorder traversal sequence of the same tree? **GATE CSE 2013**

A 10,20,15,23,25,35,42,39,30

B 15,10,25,23,20,42,35,39,30

C 5,20,10,23,25,42,35,39,30

D 15,10,23,25,20,35,42,39,30

Answer: (D) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.40 The following C function takes a simply-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

GATE CSE 2010

```
typedef struct node {
    int value;
    struct node *next;
} Node;
```

```
Node *move_to_front(Node *head) {
    Node *p, *q;
    if ((head == NULL) || (head->next == NULL)) return head;
    q = NULL; p = head;
    while (p->next != NULL) {
        q = p;
        p = p->next;
    }
    return head;
}
```

Choose the correct alternative to replace the blank line.

- ☐ A q = NULL; p->next = head; head = p;
- ☐ B q->next = NULL; head = p; p->next = head;
- ☐ C head = p; p->next = q; q->next = NULL;
- ☐ D q->next = NULL; p->next = head; head = p;

Answer: (D) q->next = NULL; p->next = head; head = p;

Q.41 The following C function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution?

```
struct node {
    int value;
    struct node *next;
};
```

GATE CSE 2008

```

Void rearrange (struct node *list ){
    struct node *p, * q;
    int temp;
    if( !list || !list-> next) return;
    p = list; q = list->next;
    while (q) {
        temp = p->value;
        p-> value = q ->value;
        q-> value = temp;
        p = q-> next;
        q = p ? p->next : 0;
    }
}

```

A 1,2,3,4,5,6,7

B 2,1,4,3,6,5,7

C 1,3,2,5,4,7,6

D 2,3,4,5,6,7,1

Answer: (B) 2, 1, 4, 3, 6, 5, 7

Q.42 The following C function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution?

```

struct node {
    int value;
    struct node *next;
};

```

GATE CSE 2005

```

Void rearrange (struct node *list ){

```

```

struct node *p, * q;

int temp;

if( !list || !list-> next) return;

p = list; q = list->next;

while (q) {

    temp = p->value;

    p-> value = q ->value;

    q-> value = temp;

    p = q-> next;

    q = p ? p->next : 0;

}

```

A 1,2,3,4,5,6,7

B 2,1,4,3,6,5,7

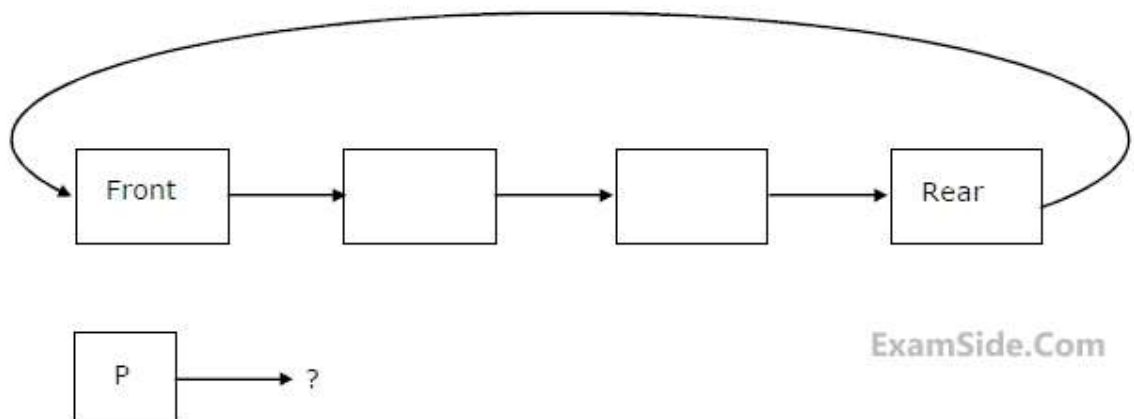
C 1,3,2,5,4,7,6

D 2,3,4,5,6,7,1

Answer: (B) 2, 1, 4, 3, 6, 5, 7

Q.43 A circularly linked list is used to represent a Queue. A single variable p is used to access the Queue. To which node should p point such that both the operations enqueue and dequeue can be performed in constant time?

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A rear node

B front node

C not possible with a single pointer

D node next to front

Answer: (A) rear node

Q.44 Consider the function f defined below.

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```
struct item {  
    int data;  
    struct item * next;  
};  
  
int f(struct item *p) {  
    return ((p == NULL) || (p->next == NULL) ||  
        ((p->data <= p->next->data) &&  
        f(p->next)));  
}
```

For a given linked list p, the function f returns 1 if and only if

A the list is empty or has exactly one element

B the elements in the list are sorted in non-decreasing order of data value

C the elements in the list are sorted in non-increasing order of data value

D not all elements in the list have the same data value

Answer: (B) the elements in the list are sorted in non-decreasing order of data value

Q.45 Linked lists are not suitable data structures of which one of the following problems?

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- ☐ A Insertion sort
- ☐ B Binary search
- ☐ C Radix sort
- ☐ D Polynomial manipulation

Answer: (B) Binary search

Q.46 In a circular linked list organization, insertion of a record involves modification of :

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- ☐ A One Pointer
- ☐ B One Pointer
- ☐ C Multiple pointers
- ☐ D No pointer

Answer: (B) One Pointer